

Siddharth Bhat

Expertise

Formal Verification (Lean) & Static Analysis (LLVM/MLIR)

- Given >4 talks @ LLVM dev meeting (Formal Semantics for LLVM & MLIR).
- Led the `lean-mlir` project for verifying optimizations in MLIR.
- Top 20 contributors to the Lean theorem prover, #2 contributor to Lean's bitvector theory.
- Coauthor of the Lean4 metaprogramming book.

Algorithms (SMT Solving, QF_BV)

- Published new, formally verified algorithms for deciding parametric bitvector theory.
- Our `bv_decide` solver placed 5th at SMT-COMP 2025 in the QF_BV division.

High Performance Computing (Loop Optimization) & Compilers (C++, LLVM, MLIR, Haskell, Lean):

- #3 Contributor to Polly, LLVM's polyhedral loop optimizer. 121 commits to Polly, 6000 LoC to LLVM, added support for Fortran GPU code generation.
- Google Summer of Code Mentor for LLVM (2016).
- #2 contributor to Asterius, a Haskell → WebAssembly compiler.
- I implemented a Haskell-style runtime on top of the WASM runtime. Contributions merged into Asterius, and eventually into GHC proper.
- Primary Author of Lean's LLVM backend.
- Papers on designing MLIR based IRs for quantum & functional compilation.

AI for Maths (Lean, F*):

- RL for proofs in the F* proof assistant @ Microsoft Research ("Towards neural synthesis for SMT-assisted proof-oriented programming" @ ICSE 2025, best paper).

Education

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| PhD in Comp. Science | University of Cambridge.
(2024 - 2026 Summer (tentative)) |
| PhD in Comp. Science | University of Edinburgh (moved to Cambridge).
(2022 - 2024) |
| Masters in Comp. Science | International Institute of Information Technology Hyderabad India.
(2020 - 2021) |
| B.Tech in Comp. Science | International Institute of Information Technology Hyderabad India.
2015 - 2020 |

Industry Research

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| Sep-Nov '24 | Amazon Web Services, Automated Reasoning Group, Austin, Research Intern.
Deciding memory (non)interference in <code>lmsym</code> , a Lean-based ARM symbolic simulator |
| Jul-Sep '23 | Microsoft Research, Redmond, Research Intern.
Retrieval Augmented theorem proving for the Fstar proof assistant. |
| May-Jul '19 | Tweag.io, Paris, France, Research Intern.
Re-implemented portions of GHC (Glasgow Haskell Compiler) runtime for Asterius (link) , a Haskell to WebAssembly compiler. Involved Haskell, C, and WebAssembly. |

Academic Internships

- Summer 2018 **ETH Zurich**, *Zurich, Switzerland*, Research Intern.
Investigating formal verification of polyhedral compilation. [PolyIR \(Link\)](#) is a formal specification of polyhedral programs.
- Summer 2018 **Google Summer of Code**, *Polly Labs*, Mentor.
Mentoring a project to enable Polly's loop optimisations into Chapel.
- Mar-Dec '17 **ETH Zurich (Scalable Parallel Computing)**, *Zurich, Switzerland*, Research Intern.
Worked on GPU code generation in Polly, a polyhedral loop optimizer for LLVM.
- May-Jul '16 **Research Intern**, *IISC Bangalore*, Bangalore.
Worked on PolyMage, DSL compiler for optimising loop transforms. Contributed to ISL and PLUTO. Implemented tiling patterns, optimised PolyMage for stencils.
- Summer 2015 **Google Summer of Code 2015**, *Google*.
Worked on VisPy, a pure Python graphics library which uses OpenGL internally for performance. Successfully completed.

Publications

First Author Papers

- Certified Decision Procedures for Width-Independent Bitvector Predicates: **Siddharth Bhat (1st)**, Léo Stefanescu, Chris Hughes, Tobias Grosser. OOPSLA 2025
- Verifying Peephole Rewriting in SSA Compiler IRs: **Siddharth Bhat (1st)**, Alex Keizer, Chris Hughes, Andres Goens, Tobias Grosser. ITP 2024
- Lambda the Ultimate SSA: **Siddharth Bhat (1st)**, Tobias Grosser. CGO 2022
- Word Embeddings as Tuples of Feature Probabilities: **Siddharth Bhat (1st)**, Alok Debnath, Souvik Banerjee, Manish Shrivastava Representation Learning for NLP, 2020

Collaborations

- Interactive Bit Vector Reasoning using Verified Bitblasting: Henrik Böving, *Siddharth Bhat*, Alex Keizer, Luisa Cicolini, Leon Frenot, Abdalrhman Mohamed, Léo Stefanescu, Harun Khan, Josh Clune, Clark Barrett, Tobias Grosser. OOPSLA 2025
- Verifying Wu's Method can Boost Symbolic AI to Rival Silver Medalists and Alpha-Geometry to Outperform Gold Medalists at IMO Geometry: Shiven Sinha, Ameya Prabhu, Ponnurangam Kumaraguru, *Siddharth Bhat*, Matthias Bethge. NeurIPS 2024 Workshop MATH-AI
- Towards Neural Synthesis for SMT-Assisted Proof-Oriented Programming: Saikat Chakraborty, Gabriel Ebner, *Siddharth Bhat*, Sarah Fakhoury, Sakina Fatima, Shuvendu Lahiri, Nikhil Swamy. ICSE 2024
- Rewriting Optimization Problems into Disciplined Convex Programming Form: Ramon Fernandez Mir, *Siddharth Bhat*, Andres Goens, Tobias Grosser. CICM 2024
- Guided Equality Saturation: Thomas Koehler, Andres Goens, *Siddharth Bhat*, Tobias Grosser, Phil Trinder, Michel Steuwer. POPL 2024
- QSSA: An SSA based IR for Quantum Computing: Anurudh Peduri, *Siddharth Bhat*, Tobias Grosser. CC 2021
- Optimizing Geometric Multigrid Computation using a DSL Approach: Vinay Vasista, Kumudha KN, *Siddharth Bhat*, Uday Bondhugula. Supercomputing (SC), Nov 2017

Open Source Contributions

- [Lean4](#) Co-developed the bitblasting theory for Lean's bitvector automation, wrote the LLVM backend for the compiler.
- [Rocq](#) Submitted issues, bug-fixes, helped improve developer documentation.
- [VE-LLVM](#) Collaboration with VE-LLVM, a formal semantics of the LLVM compiler toolchain in Coq

Polly	Implementing support for Fortran, added unified memory abilities to the CUDA backend within Polly, a polyhedral loop optimiser for LLVM. (Link to commits)
Symengine.hs	GSoC 2016. Haskell bindings to SymEngine, a C++ symbolic manipulation library.
VisPy	GSoC 2015. Rewrote scene graph for performance. Added visuals, high level API for easy use of plotting. Implemented auto-resizing with Cassowary , a linear optimisation library.
Rust	Contributed to the Rust compiler and ecosystem. Found compiler errors, fixed libraries. Was part of <i>Piston</i> , group of Rust programmers who experimented with writing game engines.
Haskell	Contributed to the Haskell ecosystem. Reported and fixed bugs in <i>stack</i> , <i>stackage</i> , <i>diagrams</i> , <i>GHC</i> , etc. (Link to GHC commits).
PLUTO	Source to Source C optimiser for loop nests. Improved the PLUTO API that had gone out of sync with master. Discovered bugs in PLUTO for diamond tiling transforms
PolyMage	DSL compiler that generates C code. Uses Polyhedral Compilation Extended the compiler to add stencils, time iterated-stencils.
PPSSPP	C++ open source PSP emulator. Wrote most of the touch handling code. Implemented atomic locks for audio performance.
Personal Projects (Compilers)	
Lean-MLIR	Formal semantics for the MLIR compiler framework, defined within the Lean4 proof assistant.
Iz	An MLIR based compiler backend for the Lean4 proof assistant.
Simplexhc	A custom compiler for a subset of Haskell. The goal is to try and apply <i>polyhedral compilation</i> ideas to compile a lazy, pure, functional programming language with LLVM as a backend. Has 64 stars on github.
Personal Projects (Formal Verification)	
Lean4 Metaprogramming Book	A textbook on metaprogramming in Lean4. I wrote the chapters on tactics and metaprogramming for embedded DSLs.
Talks & Presentations	
Barvinok	Talk at ETH Zurich: Slides describing the Barvinok algorithm to count lattice points in polyhedra
FunctionalConf '19	Talk on implementing embedded probabilistic programming languages in Haskell (Slides)
Haskell Exchange 2020	Talk on optimizing <code>smallpt-hs</code> (a port of a raytracer to haskell) to beat C++ performance (Slides)
Euro LLVM Dev 2025	How to trust your peephole rewrites: automatically verifying them for arbitrary width!
US LLVM Dev 2024	lean-mlir: A workbench for formally verifying peephole optimizations in MLIR.
US LLVM Dev 2023	(Correctly) Extending dominance to MLIR Regions.
US LLVM Dev 2023	MLIR Side Effect Modelling.
Euro LLVM Dev 2022	MLIR for Functional Programming.
FPIIndia 2021	Equality Saturation.
Functiona Conf 2019	Monad-bayes: Probabilistic programming in Haskell.